Allan Blyth

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Profile

Gameplay programmer with a 2:1 in Games Development, specialising in enemy AI and multi-phase boss systems in Unity/C#. Co-founder of Marmot Studios Ltd, taking an isometric action roguelike to Steam. Also experienced in C++/OpenGL engine programming, with a track record of delivering under tight deadlines, from 48-hour game jams to a client project.

Skills

- Languages: C#, C++
- Engines and graphics: Unity, OpenGL
- Gameplay and AI: state machines, coroutine-based timing, weighted randomness, enemy and boss behaviour, attack systems
- Other: Git, JSON, Gemini API integration, WebGL builds, controller input

Experience

Co-Founder & Programmer | Marmot Studios Ltd | 2026-Present

- Co-founded an independent game company with a five-person team, developing Kyomen, an isometric action roguelike in Unity.
- Implemented gameplay systems, enemy behaviours and boss mechanics, contributing to a playable demo.
- Handled early-stage company and publishing preparation, including Steamworks setup ahead of Steam release.

Projects

Kyomen (Unity, C#)

Created for Global Game Jam 2026; now in development for Steam release.

- Designed and implemented a four-phase boss fight, built on state machines with weighted attack selection and coroutine-based timing.
- Implemented enemy AI and attack behaviours for the core combat loop.
- Iterated rapidly from jam prototype to a playable demo.

Ballad of Auld Dunrod (Unity, C#)

Client project for a comic book publisher.

- Led programming on a delivered client project.
- Built enemy behaviours and core gameplay for WebGL on mobile browsers, working within web and mobile performance constraints.

Tower of Tales (Unity, C#)

- Created the boss's attack patterns and reactive decision-making.
- Implemented player movement and added controller support.

Voxel World Prototype (C++, OpenGL)

- Built an infinite procedural world using a chunk-based system.
- Implemented real-time directional sun lighting with voxel-cast shadows.
- Developed an interactive voxel engine with dynamic block modification and AABB collision.

AI Recap System (Unity, C#)

- Integrated the Gemini API into Unity to generate real-time, dynamic recaps of play sessions.
- Designed structured JSON built from in-game data to drive the interaction.

Other Projects

- Bob n' Steve (Unity, C#): designed and implemented the core shared-string mechanic, multiple weapon systems, and contributed to in-game voice communication.
- Global Game Jam 2025, Chill Fishing Game (Unity, C#): built core fishing mechanics and environmental interactions in a 48-hour team jam.
- Asteroids Prototype (C++, OpenGL): developed the rendering systems for a PS2-inspired visual style, with procedural asteroid generation and an explosion effects system.

Education

Glasgow Caledonian University, 2022 to 2026 BSc (Hons) Games Development, Upper Second Class Honours (2:1)